Predicting sports performance of elite female football players through smart wearable measurement platform

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Abstract

Recent development of information technology and wearable devices has led to the analysis of multidimensional sports information and the enhancement of athletes' sports performance convenient and potentially more efficient. In this study, we present a novel data platform tailored for capturing athletes' cognitive, physiological, and body composition data. This platform incorporates diverse visualization modes, enabling athletes and coaches to access data seamlessly. Fourteen elite female football players (average age = 20.6 ± 1.3 years; 3 forwards, 5 midfielders, 4 defenders, and 2 goalkeepers) were recruited from National Taiwan Normal University, Taiwan, as the primary observational group, and 12 female university students without regular sport/exercise habits (average age = 21.6 ± 1.3 years) were recruited as control group. Through multidimensional data analysis, we identified significant differences in limb muscle mass and several cognitive function scores (e.g. reaction times of attention and working memory) between elite female football varsity team and general female university students. Furthermore, 1-month heart rate data obtained from wearable devices revealed a significant negative correlation between average heart rate median and cognitive function scores. Overall, this study demonstrates the potential of this platform as an efficient multidimensional data collection and analysis platform. Therefore, valuable insights between cognitive functions, physiological signals and body composition can be obtained via this multidimensional platform for facilitating sports performance.

KeywordsElite female football athletes; Sports big data platform; Wearable device; Cognitive function; Body composition

1. Introduction

The immense global popularity of football in recent decades has established it as one of the most fiercely competitive sports (Haugaasen and Jordet, 2012). Concurrently, nurturing exceptional football players has evolved into a lucrative and prestigious enterprise for clubs and national associations (Relvas et al., 2010). Due to the imperative need for talent identification, professionals in this domain continually grapple with the question of which methods are most effective for this process. Previous studies have shown that potential football players can be identified based on anthropometric, psychological, physiological, and skill factors. The development of objective assessment tools to quantify strategic and methodological competence could improve talent selection process (Hoare and Warr, 2000;(Hoare and Warr, 2000) Williams and Reilly, 2000). Despite the longstanding advocacy for multidimensional approaches to talent identification (Abbott et al., 2005; Vaeyens et al., 2008; Williams and Reilly, 2000), many clubs and associations still predominantly rely on subjective data derived from coach assessments (Christensen, 2009; Larkin and Reeves, 2018). However, amidst this traditional approach, the advent of the big data era has sparked a significant transformation in the landscape of talent identification and sports analytics.

Big data informatics are critical for athletes because they provide valuable information about exercise performance, health data, training statistics, and analysis, which assist them improve their training and gaming strategies. They have become an indispensable asset for winning competitions (Gudmundsson and Horton, 2017). Advanced big data collection and analytical techniques have precipitated a paradigm shift within the sports industry. The vast amount of sports data available presents exciting opportunities and complex challenges in the growing field of sports big data analytics (Pappalardo et al., 2019). The concept of big data, as articulated by the McKinsey Global Institute, encompasses four essential attributes: volume, variety, velocity, and value (Gobble, 2013). This shift toward data-driven decision-making marks a crucial turning point in the realm of talent identification and player development within football and beyond.

As stated above, the purpose of this study is to provide a data platform for tracking physiological and psychological states of female football athletes, as well as to provide coaches with an effective means of monitoring their members' states during training and competition. In addition, we use the platform to investigate the relationship between cognitive function, physical fitness, and physiological status among female football participants, with the goal of predicting their athletic performance. We hypothesize that female football athletes have higher cognitive performance than control group, interacted with physical fitness and attributed to long-term training benefits. On the other hand, we also hypothesize that female football athletes' cognitive performance can be reflected with some physiological markers.

This study has developed a data platform tailored for elite female football players and coaches. It integrates data collected from Garmin wearable devices and cognitive performance data from "The Brain Gym" (Chen et al., 2024). The data platform employs a MySQL database and various frontend technologies, including Web, Android APP, and iOS APP. To ensure the flexibility for data expansion and continuous network services, it is hosted on cloud-based virtual servers. The present study will introduce the data sources (Garmin wearable devices, “The Brain Gym” app, and Tanita body composition device) and their respective data content and their relevance to sports physiology. Additionally, it will outline the data collection and storage planning (Ubuntu operating system, MySQL database). Furthermore, it will describe the user interface data input functionality and basic data display and analysis features.

For athletes, by analyzing this data, we display athletes' individual physiological summaries, cognitive function data, and other information on a website and a smartphone app to enhance athletes' self-regulation in sports. For coaches, we have designed a visual interface that allows them to access the entire team's physical training records. Coaches can query physiological data on various aspects such as training, sleep, and stress for each player. In addition to data visualization, this data platform also offers basic machine learning analysis capabilities. Machine learning can provide predictive insights into potential injury occurrences. Coaches can use athletes' biometric data to adjust training intensities. This can provide with precise sports forecasting and prevents potential injuries.

2. Management of multiple data sources

In sports, a combination of physiological abilities like anaerobic capacity, psychological traits such as self-efficacy, and specialized competencies including technical and tactical skills are essential for superior performance (Sarmento et al., 2018). Moreover, cognitive functions and abilities are proposed as contributing factors to exceptional athletic achievement (Scharfen and Memmert, 2019). An athlete's physical profile consists of their ever-changing cognitive and psychological traits, as well as their physiological characteristics during training and recovery. To bridge these aspects, digital recording of athletes' physiological features facilitates comprehensive analysis, enabling the detection of physiological changes resulting from training. Therefore, for the effective recording of athletes' psychological and physiological characteristics, the use of mobile and wearable devices is essential.

Recent advancements in wearable technology such as Garmin watches, it has been possible to quantitatively record athletes' physiological data (Seshadri et al., 2019). Due to Garmin's data integration APIs (Application Programming Interfaces), developers can access measurement data from Garmin wearable devices. After comparing various models of Garmin wearable devices and considering factors such as data quality, portability, and battery life, this study chose the Garmin Forerunner 255 to measure various physiological data of athletes. The Garmin Forerunner 255 can measure physiological data, such as HRV (Heart Rate Variability), blood oxygen levels, heart rate, and step count. It also offers algorithm-based data estimations such as sleep status, psychological stress, body battery, and physiological data sets for different states (e.g., sleep and exercise). The data can be obtained through the Garmin APIs.

Our laboratory created an app named “The Brain Gym” to gather cognitive function data, such as reaction time, attention, working memory capacity, and geometric shape planning ability etc. (Chen et al., 2024). It is designed to run on tablets and quantitatively measure cognitive functions through interactive gaming. “The Brain Gym” app employs game-based tasks to measure participants' reaction times (in milliseconds), memory capacity (in unit count), and geometric pattern logical reasoning ability (accuracy rate). Utilizing tablets as the platform not only allows participants to engage in cognitive testing without the need for a computer but also ensures a highly consistent human-machine interface condition for the test takers.

The data processing workflow can be summarized as: (1) signal acquisition, (2) information visualization, and (3) data analysis. This section primarily focuses on signal acquisition. We utilize Garmin wearable devices, “The Brain Gym” app, and Tanita to capture various data from athletes, including physiological and cognitive functions. The data collected in this study are shown in Table 1.

The diverse dataset mentioned above encompasses physiological data from Garmin wearable devices, cognitive function scores from “The Brain Gym” app and body composition from Tanita. The data is stored in MySQL database on Vultr cloud servers, with geographical redundancy, mirrored backups, and high-capacity storage in Tokyo and New York. The backend server is built on the Python Flask framework and served using the Nginx server for web services. Cloudflare is employed for DNS redirection to mitigate DDoS attacks. SSH login requires two-factor authentication, and regular operating system patching is performed to maintain system security as much as possible.

The following will outline the physiological data collected and recorded from Garmin wearable devices, the cognitive function scores

Table 1 Types of data recorded in the study.

Source	Type (unit)	Directions
Garmin forerunner 255	Heart rate (beats per minute, bpm)	The frequency of heart beats per minute.
	Body battery (N/A)	The Garmin Body Battery value (ranging from 0 to 100) is calculated based on data such as sleep, HRV (Heart Rate Variability), activity, and stress levels.
	Stress (N/A)	The Garmin Stress value (ranging from 0 to 100) increases the activation level of the body when sympathetic activity dominates the ANS and parasympathetic activation is low.
	Steps (counts)	The individual movements in walking.
	Floors (counts)	The number of floors climbed or descended.
	HRV (ms for SDNN and RMSSD)	HRV stands for Heart Rate Variability. The variation in the time interval between consecutive heartbeats provides insights into the function of the autonomic nervous system. HRV can be calculated in various ways, and Garmin's Health Snapshot feature utilizes RMSSD and SDNN to calculate HRV over a 2-min period. Additionally, Garmin employs the RMSSD algorithm to calculate HRV every 5 min during the sleep process.
	SpO_2 (%)	SpO_2 stands for peripheral capillary oxygen saturation, which measures the percentage of oxygen bound to hemoglobin in the blood.
	VO_2 max (mL)	VO_2 max refers to the maximum amount of oxygen in milliliters that you can consume per minute per kilogram of body weight at your peak performance. It serves as an indicator of aerobic fitness and increases as your fitness level improves.
	Respiration (Brpm)	Respiration measures breathing rate per minute.

Source	Type (unit)	Directions
	Sleep (N/A)	Sleep summary represents the duration of sleep and automatically categorized sleep stages (such as light sleep, deep sleep) during the sleep period of the user. It is based on the sleep records generated from data collected by the user's device.
	Activity (N/A)	The activity summary provides advanced information about discrete fitness activities (such as running or swimming). These summaries are generated when users actively record activities on their devices.
The brain gym	Reaction time (ms)	Reaction time measures the time it takes for participants to perceive the change from blue to red of a circular icon displayed on the screen of the 7th-generation Apple iPad using "The Brain Gym" app.
	Movement time (ms)	Movement time measures the duration it takes for participants to move their finger from the bottom of the screen to the top of the screen on the 7th-generation Apple iPad while using "The Brain Gym" app.
Tanita MC 980 MA PLUS	BMI (kg/m^2)	
	BMR (kcal)	Basal metabolic rate, the number of calories which body needs to maintain basic physiological functions while at rest.
	Fat (%)	Percentage of body weight that is comprised of fat tissue. It indicates the proportion of fat mass relative to total body weight.
	Fat mass (kg)	The total amount of fat tissue in the body.
	Muscle mass(kg)	The total amount of muscle tissue in the body. It includes skeletal muscles and other muscle tissues.
	RA muscle mass (kg)	Right arm muscle mass, specifically measuring the muscle mass of the right arm.
	RA fat mass (kg)	Right arm fat mass, specifically measuring the fat mass of the right arm.

Source	Type (unit)	Directions
	RA fat (%)	Percentage of fat tissue in the right arm.
	LA muscle mass (kg)	Left arm muscle mass, specifically measuring the muscle mass of the left arm.
	LA fat mass (kg)	Left Arm fat mass, specifically measuring the fat mass of the left arm.
	LA fat (%)	Percentage of fat tissue in the left arm.
	RL muscle mass (kg)	Right Leg muscle mass, specifically measuring the muscle mass of the right leg.
	RL fat mass (kg)	Right leg fat mass, specifically measuring the fat mass of the right leg.
	RL fat (%)	Percentage of fat tissue in the right leg.
	LL muscle mass (kg)	Left leg muscle mass, specifically measuring the muscle mass of the left leg.
	LL fat mass (kg)	Left leg fat mass, specifically measuring the fat mass of the left leg.
	LL fat (%)	Percentage of fat tissue in the left leg.

from "The Brain Gym" app and body composition from Tanita in this study.

2.1 Garmin wearable devices

2.1.1 Heart rate

Heart rate shows the frequency of heart beats per minute could be measured by photoplethysmography (PPG). PPG is a noninvasive optical technique used to monitor both heart rate (HR) and beat-to-beat interval (BBI). It integrates a light emitter and a photodetector to measure the volumetric variations of blood flow according to Beer's law (Castaneda et al., 2018). The heart rate data from the Garmin API comes from daily records, activity logs, and Health Snapshot. Daily record data is recorded every 15 s and calculates the day's maximum, minimum, and average heart rate. During activity, heart rate is recorded approximately every 1–4 s. The heart rate is recorded every 1 s, totaling 2 min in Health Snapshot mode.

Postexercise heart rate recovery (HRR) serves as a reliable measure reflecting the balance between sympathetic and vagal activity. Additionally, it plays a significant role in prescribing and monitoring athletic training programs. It could be a useful tool to track changes in training status in athletes (Daanen et al., 2012).

2.1.2 Heart rate variability (HRV)

Heart rate variability (HRV) has been utilized as a biomarker of mental and physical stress (Castaldo et al., 2015; Thayer et al., 2012). The HRV is the variations in the duration of beat-to-beat intervals (BBIs) (Sacha, 2014) measured by PPG. Research showed that HRV could reliably be estimated by PPG signals (Choi and Shin, 2017). Both heart rate and HRV reflect the balance between the parasympathetic and sympathetic nervous systems, where the parasympathetic nervous system predominates during "rest" while the sympathetic nervous system dominates during stress (Castaldo et al., 2015; Thayer et al., 2012; Umetani et al., 1998). For healthy individuals not frequently exposed to stress, higher resting HRV is associated with better health outcomes, including lower mortality rates, reduced depression, improved sleep, and greater cognitive flexibility (Carnevali et al., 2018; Castaldo et al., 2015; Gouin et al., 2015; Ottaviani et al., 2015; Pradhapan et al., 2014; Thayer et al., 2012).

The calculation of HRV is broadly categorized into two main types: time domain analysis and frequency domain analysis (Rajendra Acharya et al., 2006). Garmin wearable devices provide HRV algorithms based on time domain analysis, including RMSSD and SDNN algorithms. In the Garmin "Health Snapshot", RMSSD and SDNN are used to calculate HRV based on the BBI signal within 2 min. In sleep mode, use RMSSD to calculate HRV and record every 5 min. Both the Health Snapshot and HRV data during sleep can be received through the Garmin API.

2.1.3 Sleep

The American Academy of Sleep Medicine, the Sleep Research Society, and the National Sleep Foundation recommend that adults get at least 7 h or more of sleep per night (Hirshkowitz et al., 2015; Watson et al., 2015). Numerous studies indicate that many athletes face challenges with obtaining sufficient sleep (Samuels, 2008). When compared to individuals who are not athletes, athletes typically sleep for shorter durations, averaging between 6.5 and 6.7 h per night, and their sleep quality tends to be poorer (Fietze et al., 2009). Research suggests that certain athletes, notably those in the National Football League (NFL), have elevated rates of obstructive sleep apnea/sleep-disordered breathing, a condition that diminishes the quality of night time sleep and frequently leads to heightened levels of daytime drowsiness (Albuquerque et al., 2010). Sleep is particularly crucial for athletes due to its significant impact on athletic performance.

The Garmin wearable device determines the user's sleep status (light, deep, REM, or awake) by analyzing heart rate variability (HRV), breathing rate, heart rate, and body movement (Saalasti and Ruhanen, 2021). The sleep data transmitted through the Garmin API includes SpO_2 and respiration frequency recorded every 60 s, the start and end times of four sleep stages (deep, light, REM, awake), along with their respective assessment scores, and an overall assessment score for the entire sleep duration. Additionally, heart rate variability (HRV) calculated using the RMSSD algorithm is recorded every 5 min.

2.1.4 Stress

Stress is the body's natural response to challenges and environmental stimuli, helping us quickly respond to situations. For the Garmin wearable device, the estimation of stress levels (ranging from 0 to 100) is conducted by the Firstbeat Analytics engine, predominantly leveraging a composite of heart rate (HR) and heart rate variability (HRV) data (Kettunen and Saalasti, 2008b). The Garmin device estimates stress every 3 min and shares the data through the Garmin API.

2.1.5 VO_2Max

VO_2Max stands for maximal oxygen consumption and represents the highest amount of oxygen in milliliters that a person can use per minute per kilogram of body weight during peak performance (Fu et al., 2023; Wang et al., 2015). Ever since Hill proposed it in 1923, maximal oxygen uptake (VO_2Max) has continued to be a globally acknowledged “gold standard” for measuring aerobic capacity and assessing cardiopulmonary function. It is crucial in various domains, including competitive sports, public fitness, and clinical medicine. This includes tasks such as selecting talented athletes, assessing motor function, developing training programs, and estimating cardiovascular health and risks. The measurement of VO_2Max encompasses both direct and indirect methods. Direct methods measure oxygen consumption during exercise using exhaled gases, while indirect methods estimate VO_2Max using factors like heart rate, distance, or time. Indirect methods include exercise tests like the

1.5-mile run, the 12-min run, and the 20-m shuttle run. They also involve prediction equations based on non-exercise factors, like the ones created by Wasserman, NASA, Jackson, and others (Brooks et al., 2005; Buttar et al., 2019). The Garmin device estimates VO_2Max daily and shares the data through the Garmin API.

Football involves a combination of both anaerobic and endurance elements. Players engage in short bursts of anaerobic activities such as sprints, quick changes in direction, and explosive movements. At the same time, they also need endurance to sustain the duration of the game and the continuous running involved. Excellent performance in endurance athletes is typically associated with factors such as running economy, anaerobic threshold, and maximum oxygen uptake (VO_2Max). Therefore, it is suggested that factors such as VO_2Max may potentially predict the distinction between elite and non-elite football players, albeit with a conservative approach (Lorenz et al., 2013).

2.1.6 Body battery
The body battery feature of your Garmin watch is designed to constantly monitor individual energy levels throughout the day. Firstbeat Analytics uses body battery energy tracking to show how physical activity, stress, rest periods, and sleep affect people's energy levels (Saalasti et al., 2016). The body battery feature uses heart rate, heart rate variability (HRV), and movement data to track energy levels. This analysis aims to recognize significant physiological states and elucidate their influence on the body's energy reserves. It basically records the user's sleep or wake patterns, detects periods of physical activity, and tracks stress levels during inactive periods. The Garmin device estimates the body battery every 3 min and records events related to the body battery, such as sleep, nap, activity, recovery, and stress. These logs are shared through the Garmin API.

2.1.7 SpO_2
The Garmin wearable device utilizes pulse oximetry to measure the reflection of red and infrared light flux from the microvasculature of the dermis layer, assessing the levels of Hb and HbO_2 and calculating SpO_2, represented as a percentage (Nitzan et al., 2014). In general, this value should typically exceed 95% across various settings; however, it may be affected by factors such as altitude, activity level, and an individual's

health status. The Garmin device records SpO_2 every minute in “All Day Mode” and shares the data through the Garmin API.

2.1.8 Respiration

Similar to heart rate, respiration rate is controlled by the autonomic nervous system (ANS). The time between heartbeats shortens slightly during inspiration, while it lengthens during exhalation, known as respiratory sinus arrhythmia (RSA). A resting adult generally has a normal respiration rate of 12–20 breaths per minute. Lower respiration rates at rest usually indicate better physical fitness. Physically fit people tend to have lower rates even when active and return to lower rates more quickly after being active (Kettunen and Saalasti, 2008a). The Garmin device records respiration frequency every minute and shares the data through the Garmin API.

2.2 “The Brain Gym” app for cognitive data

“The Brain Gym” app comprises a series of cognitive tasks to evaluate several cognitive function and capacity. The internal programming for the mobile platform app prototype was carried out with the Apple iOS and Android application (Fig. 1). These tasks are based on conventional cognitive assessments, allowing for the straightforward and practical evaluation of cognitive function performance across diverse demographics. Each assesses performance in cognitive functions associated with reaction performance, working memory, inhibitory control, processing speed, decision making, and executive functions (Chen et al., 2024), which has been shown to important for individual difference associated with sports expertise (Wang et al., 2013, 2017, 2020a,b, 2023). On the other hand, the data collection in this app includes reaction time, accuracy, memory capacity, information integration ability, and response performance. This app allows for the fast testing of cognitive performance in a variety of population groupings. Furthermore, it also broadens the scope of cognitive function training to athletes of all ages, improving the efficacy of cognitive function testing and training.

In this study, we adopted the tasks from Chen and colleagues (Chen et al., 2024) using the simple reaction task, sequential memory test, and path-tracking task to assess the cognitive ability of female athletes (Fig. 2). A simple reaction task assesses reaction time during the preparation phase and response speed for movement. On the other hand, sequential memory tasks can assess female football athletes' working memory ca-

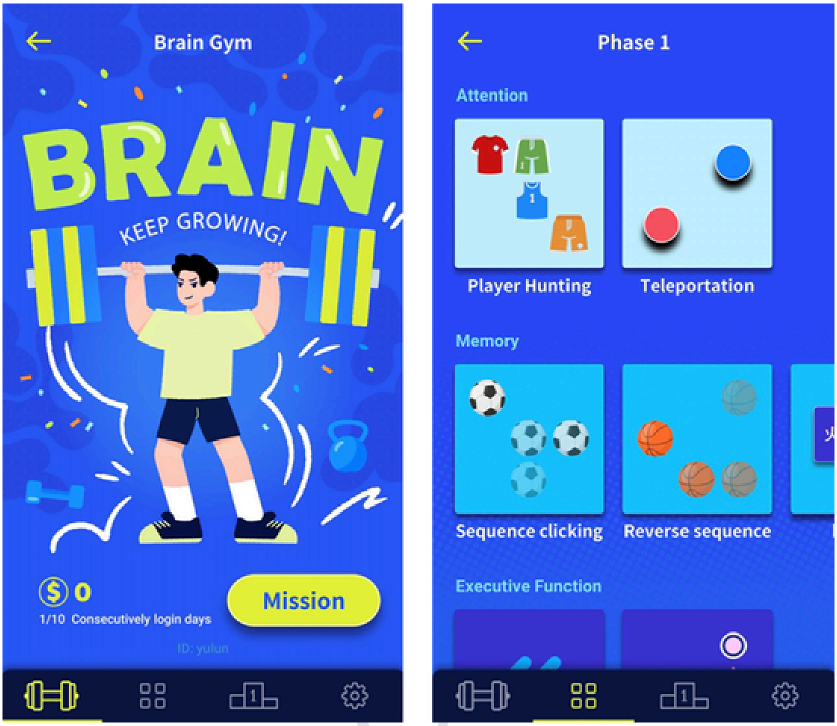
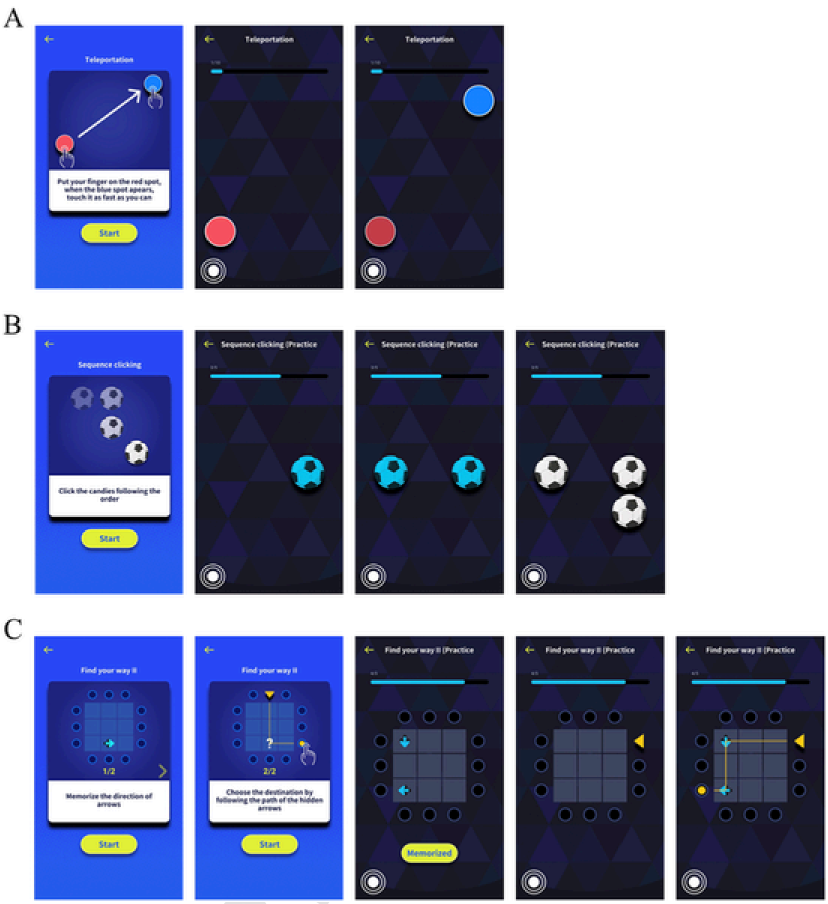


Fig. 1 “The Brain Gym” app. All tasks in this app were selected based on the conventional psychology experiments. This app can measure the cognitive function regarding processing speed, working memory, executive function, inhibitory control, and decision making.

pacity. The path-tracking task assessed executive function by measuring response accuracy, memory capacity, and processing speed. Data collected by the app will be automatically transferred to a data platform for further analysis, and the results will provide assessments of athletes' brain behavior and cognitive function-related performance. Additionally, it can also be combined with physiological data that has been gathered to do extensive studies.

2.3 Body composition

Tanita MC 980 MA PLUS (Tanita, Tokyo, Japan) used a bioelectrical impedance analysis (BIA) method to determine the body composition. The device operates in the following impedance (resistance and reactance) frequencies: 1, 5, 50, 250, 500, and 1000 kHz. Participants held both



hands on the sensor and stood straight on the platform. Each participant used an 8-contact electrode system to measure BMI (kg/m^2), BMR (kcal), total body water (TBW), fat (%), fat mass (kg/m^2), muscle mass (kg), and indexes linked to the trunk and limb.

3. System design

This section explores the multifaceted dimensions associated with the system's conceptualization and structural framework. This comprehensive examination encompasses critical elements, including the Backend, Frontend, and Database.

3.1 Backend

In the backend architecture of the sports big data platform, we adopt a decoupled development approach, separating the frontend and backend functionalities. The backend is developed using the Flask framework in Python, renowned for its simplicity and flexibility. Flask facilitates the creation of web applications through its lightweight and modular design, allowing developers to build scalable and efficient backend systems. Meanwhile, the frontend architecture comprises the rendering of HTML and CSS templates using Jinja2, a template engine for Python. Jinja2 enables the generation of dynamic content based on data from the backend, enhancing the user experience with personalized and interactive elements. Additionally, JavaScript is employed to facilitate user interactions and dynamic content updates, further enriching the frontend functionality. This architectural setup ensures the effective separation of concerns between the frontend and backend components, promoting modularity, maintainability, and scalability within the sports big data platform.

Using Python in development allows developers to use many third-party libraries to improve development efficiency. For instance, developers use NumPy to perform matrix computations and SciPy to perform digital filtering for the pre-processing of HRV calculation for ECG signals. Additionally, Matplotlib serves as a tool for data visualization, including correlation heatmaps. Regarding information security, Authlib is utilized for data security authentication for Garmin API.

Fig. 2 The simple reaction task, sequential memory task, and path-tracking task in "The Brain Gym" app. (A) The procedure for a simple reaction task. To begin a trial, participants touch and hold the preparing cue (a red dot). After a random duration (3–5 s), a target (a blue dot) will appear. Participants have to lift their index finger and tap the target as fast as possible. (B) At the beginning of the trial, the ball stimuli were displayed randomly on the screen. Participants were required to memorize the location and order of presentation of the ball stimuli. Participants must tap the ball stimuli at the corresponding areas in sequence. The set size will increase following predetermined trials. (C) Participants were instructed to memorize the position and direction of each arrow. When the participants touch the start answer button (which says "Memorized"), the arrows disappear, and a triangle's starting point appears simultaneously in one of the 12 spots surrounding the grid. Participants were asked to respond to the path's endpoint according to the direction of the disappearing arrows.

We can also integrate natural language processing tools like BERT-based models or OpenAI's ChatGPT API into sports big data platforms using third-party Python modules. Furthermore, the Python developer community offers a multitude of third-party modules for use with the Flask framework. For instance, Flask_cors facilitates cross-origin resource sharing (CORS), while Flask_socketio enables web sockets implementation for real-time information exchange among multiple users on the platform.

In Flask, the Blueprint class helps in dividing a web application into different modules. Each module can have its own routes, view functions, and features. In our platform, we define separate Blueprints for distinct functionalities, such as "Sport" and "MachineLearning," which not only differentiate the features but also alleviate the complexity of development and maintenance.

3.2 Frontend

We utilized HTML, CSS, JavaScript, jQuery, and the Bootstrap framework to construct the frontend interface. The incorporation of jQuery and Bootstrap frameworks significantly streamlines the frontend development process while providing extensive UI components and elements, thereby expediting development and enhancing visual design quality. Employing the principles of Responsive Web Design (RWD), the frontend interface is designed to be responsive to various screen widths, thereby enhancing the readability of the sports big data platform.

To dynamically visualize and interact with athlete data such as sleep patterns and HRV, we utilized the Chart.js JavaScript library for development. As shown in Fig. 3, cognitive performance scores of athletes are displayed using line charts. Fig. 4 illustrates the distribution of athlete data including sleep, HRV, psychological stress, and Body Battery over the past 2 weeks. Fig. 5 demonstrates the dynamic presentation of sleep patterns, including deep, light, REM, and awake stages, as well as SpO_2 variations throughout the night, upon user selection of a specific date from the sleep bar chart. The data demonstrate sleep quality and assist athletes in gaining deeper insights into their sleep processes. Fig. 6 shows the information handling process for a user to access their personal dashboard.

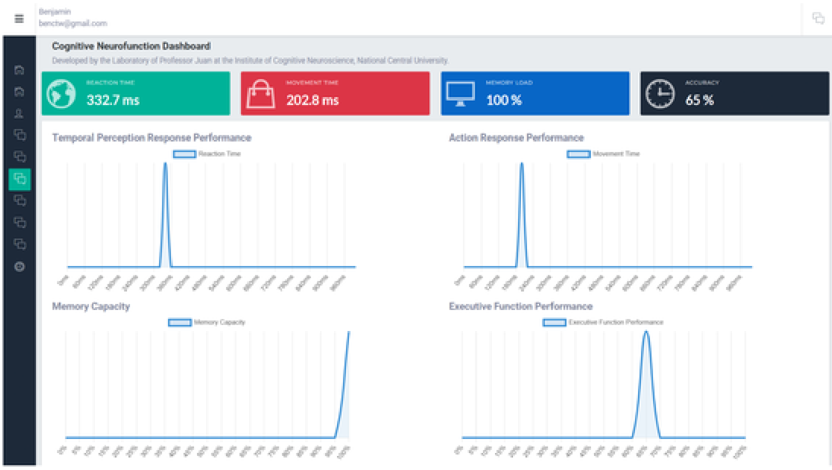


Fig. 3 Cognitive performance scores.

Fig. 4 Distribution of personal data in 2 weeks.

3.3 Database

Considering the reliability of database applications, we have chosen MySQL as the database for the sports big data platform. The decision to use MySQL for storing Garmin's multi-user fitness and health data is based on several key factors. MySQL is renowned for its robustness and maturity in data storage and management, making it a dependable choice. The MySQL developer community offers abundant resources for technical support during the development and maintenance processes. It provides support for transactions, SQL functionalities, indexing, and query

Fig. 5 Visualization of sleep patterns and SpO_2 variation.

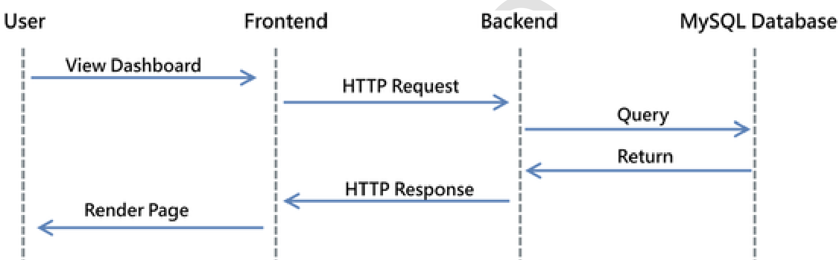


Fig. 6 Request diagram.

optimization. MySQL can handle diverse datasets, including multi-user fitness and health data from Garmin's API. Although not as scalable as some NoSQL databases, MySQL still delivers commendable performance and scalability capabilities when appropriately configured and optimized.

4. Data visualization

In this section, we provide an overview of the user interfaces for athletes and coaches, highlighting the data available to each.

4.1 Athlete interface

The athlete interface design emphasizes helping athletes track their training and recovery status, focusing on convenience, readability, and security. The sports big data platform is accessible on both computers and mobile devices, allowing athletes to easily monitor their training progress anytime using their own mobile devices. Chart.js is a JavaScript library that offers interactive charts for athletes to view key information about their physical recovery. It makes this information easy to understand. The athlete interface provides access to several dashboards:

4.1.1 Garmin dashboard

The Garmin dashboard displays data such as sleep, HRV, stress levels, and Body Battery over time in bar charts as Fig. 4 shows. Users can click on specific data points within the bar chart to view detailed information. For instance, clicking on the sleep bar chart reveals details of sleep stages (deep, light, REM, awake) and SpO_2 (blood oxygen saturation) data changes as Fig. 5 shows.

When athletes click on the data for HRV, psychological stress, or Body Battery for a specific day, it will display a detailed time-based distribution chart for that day, as shown in Figs. 7–9.

4.1.2 Cognitive dashboard

The cognitive dashboard visualizes the data measured by "The Brain Gym" app and shows information about time perception, reaction time, memory capacity, and logical execution performance, as Fig. 3 shows.

4.2 Coach interface

The coach interface enables coaches to efficiently oversee the training and physical recovery status of the entire team while also monitoring

Fig. 7 Daily HRV distribution chart.

Fig. 8 Daily stress distribution chart.

Fig. 9 Daily body battery distribution chart.

their cognitive function performance. The interface is divided into three sections: left column, center column, and right column as Fig. 10 shows.

4.2.1 Left column

The left column allows coaches to switch between different teams. When switching teams, the player list in the center column updates accordingly.

4.2.2 Center column

The center column displays a list of players. Coaches can click on individual players to access their training and physical recovery status in the right column.

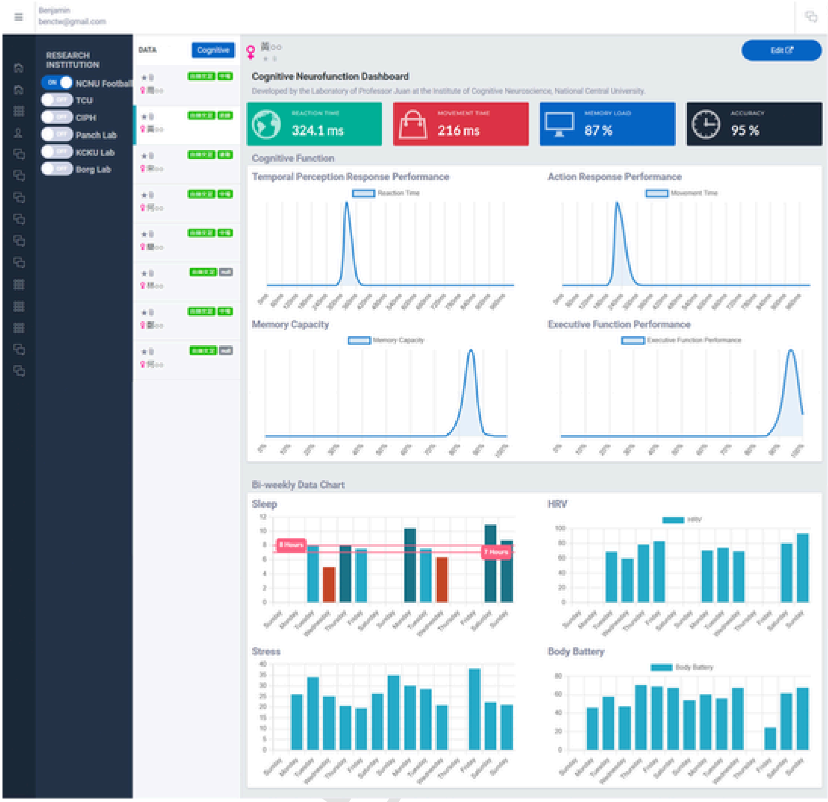


Fig. 10 Coach interface.

4.2.3 Right column

The right column provides detailed information on individual players' training and physical recovery status. Coaches can also view cognitive function scores, categorized by player position (goalkeeper, forward, midfielder, defender), to assess the distribution of cognitive function scores among players of the same position as Fig. 11 shows.

5. Data analysis

Fourteen young female football players (average age = 20.6 ± 1.3 years; 3 forwards, 5 midfielders, 4 defenders, and 2 goalkeepers) were recruited from National Taiwan Normal University, Taiwan, as the primary observational group, and 12 female university

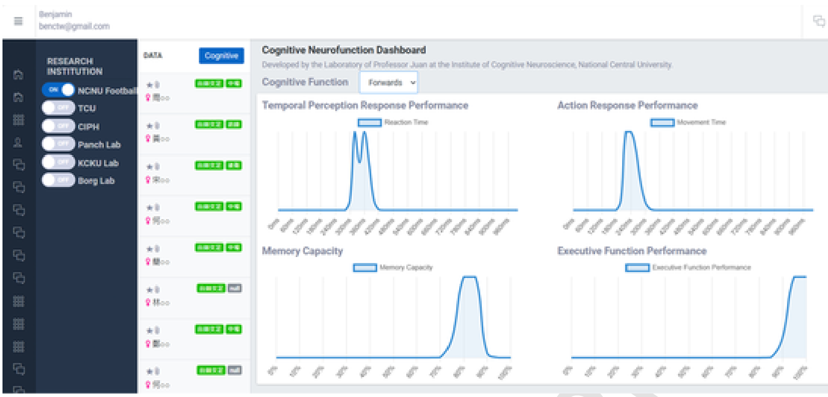


Fig. 11 Cognitive function score charts according to different field positions.

students without regular sport/exercise habits (average age = 21.6 ± 1.3 years) were recruited as control group. The 14 female football players were playing at the top level of the University Football Association. No one reported having a history of neurological or cardiovascular disorders or taking medications that impaired cognitive processes, and all participants had normal or corrected-to-normal vision. The current study followed the National Taiwan University's Ethical Principles and Regulations for Social and Behavioral Research (Ref: 202107EM002). Informed consent was provided by each participant before experiments.

5.1 Participant demographics

Table 2 shows participant demographic data and their variables of body composition. Demographic variables including age, height, fat, fat mass, RA fat mass, LA fat mass, RL fat, and LL fat, did not differ between groups (age, [$F(1, 24) = 3.26$, $P = 0.084$, $\eta^2 = 0.119$]; height, [$F(1, 24) = 0.203$, $P = 0.657$, $\eta^2 = 0.008$]; fat, [$F(1, 24) = 1.86$, $P = 0.185$, $\eta^2 = 0.072$]; fat mass, [$F(1, 24) = 0.169$, $P = 0.685$, $\eta^2 = 0.007$]; RA fat mass, [$F(1, 24) = 0.509$, $P = 0.482$, $\eta^2 = 0.021$]; LA fat mass, [$F(1, 24) = 0.078$, $P = 0.782$, $\eta^2 = 0.003$]; RL fat, [$F(1, 24) = 0.004$, $P = 0.951$, $\eta^2 < 0.001$]; and LL fat, [$F(1, 24) = 0.012$, $P = 0.915$, $\eta^2 < 0.001$]). In contrast, there was a significant group difference including weight, BMI, BMR, muscle mass, RA muscle mass, RA fat, LA muscle mass, LA fat, RL muscle mass, RL fat mass, LL muscle mass, and LL fat mass (weight, [$F(1, 24) = 6.44$, $P = 0.018$, $\eta^2 = 0.211$]; BMI, [F (1,

Table 2 Demographics of participants in each group.

Group	Female football players (n = 14)		Control group (n = 12)	
	Mean	SD	Mean	SD
Age (years)	20.6	1.3	21.6	1.3
Height (cm)	162.4	5.1	161.3	7.4
Weight (kg)	59.2*	6.9	51.7	8.2
BMI (kg/m^2)	22.4**	2.1	19.9	2.6
BMR (kcal)	1332.6***	117.8	1135.9	122.5
Fat (%)	27	3.7	29.4	5.1
Fat mass (kg)	16.2	3.8	15.4	5.2
Muscle mass(kg)	40.4***	3.4	34.2	3.6
RA muscle mass (kg)	1.9***	0.2	1.4	0.2
RA fat mass (kg)	0.6	0.1	0.5	0.2
RA fat (%)	21.9*	3.6	26	5.6
LA muscle mass (kg)	1.8***	0.2	1.4	0.2
LA fat mass (kg)	0.6	0.1	0.5	0.2
LA fat (%)	22.6*	3.4	26.3	5.7
RL muscle mass (kg)	8.4***	0.8	6.6	0.8
RL fat mass (kg)	3.9*	0.9	3	0.8
RL fat (%)	29.8	3.4	29.7	4.4
LL muscle mass (kg)	8.3***	0.9	6.5	0.8
LL fat mass (kg)	3.9*	0.9	3	0.8
LL fat (%)	30.3	3.4	30.1	4

SD = standard deviation; BMI = body mass index; BMR = basal metabolic rate; RA = right arm; LA = left arm; RL = right leg; LL = left leg. ***$P < 0.001$, **$P < 0.01$, *$P < 0.05$.

24) = 7.50, $P = 0.011$, $\eta^2 = 0.238$]; BMR, [F (1, 24) = 17.36, $P < 0.001$, $\eta^2 = 0.420$]; muscle mass, [F (1, 24) = 20.12, $P < 0.001$, $\eta^2 = 0.456$]; RA muscle mass, [F (1, 24) = 34.16, $P < 0.001$, $\eta^2 = 0.587$]; RA fat, [F (1, 24) = 4.87, $P = 0.037$, $\eta^2 = 0.169$]; LA muscle mass, [F (1, 24) = 24.85, $P < 0.001$, $\eta^2 = 0.509$]; LA fat, [F (1, 24) = 4.31, $P = 0.049$, $\eta^2 = 0.152$]; RL muscle mass, [F (1, 24) = 29.41, $P < 0.001$, $\eta^2 = 0.551$]; RL fat mass, [F (1, 24) = 6.95, $P = 0.014$, $\eta^2 = 0.224$]; LL muscle mass, [F (1, 24) = 25.80, $P < 0.001$, $\eta^2 = 0.518$]; and LL fat mass [F (1, 24) = 6.76, $P = 0.016$, $\eta^2 = 0.220$]). Interestingly, the results revealed that the muscle mass of the limbs in female football players was significantly higher than the control group. This may be attributed to the long-term effects of regular exercise training. Therefore, this study uses total body muscle mass index as the independent variable for the following results discussion.

5.2 Cognitive function and body composition

This study uses data from the intelligent platform to examine the relationship between cognitive function performance and body composition among female football players and a control group. In cognitive function testing, movement time, referring to the duration taken to complete a particular action or task, is a critical measure of motor control and coordination. Fig. 12B shows an apparent negative correlation between movement time and muscle mass when combining female football players and sedentary control groups ($r = -0.4$, $P = 0.041$), indicating that movement time tends to decrease with muscle mass. Further comparison of the differences between the two groups revealed that the muscle mass of female football players was substantially higher than the control group, [t $(24) = 4.47$, $P < 0.001$, $d = -1.764$] (see Fig. 12A). It was also evident

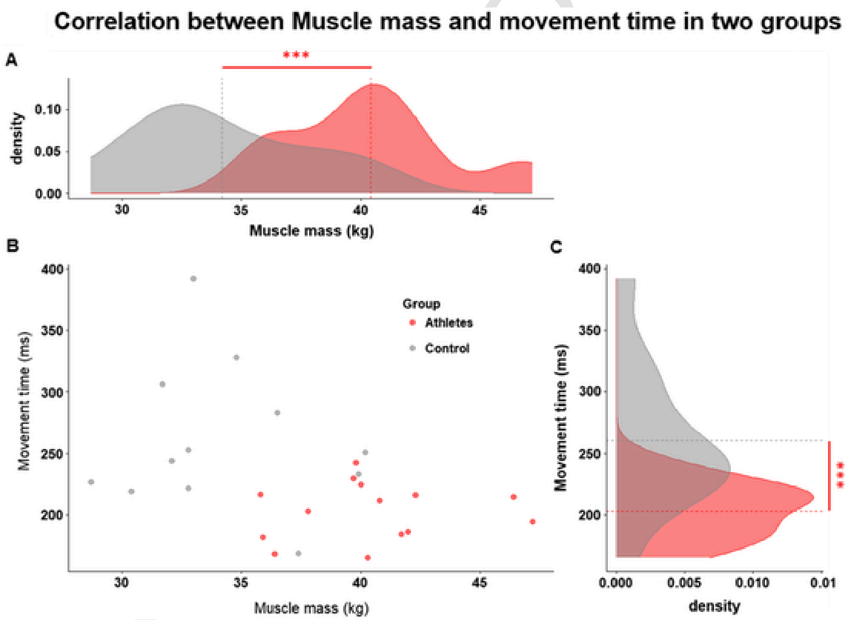


Fig. 12 The correlation between muscle mass and movement time: (A) The distribution of muscle between the female football players and the control group. (B) The correlation between the muscle mass and movement time in the two groups. (C) The distribution of movement time performance between the female football players and the control group; the significant differences were seen in both muscle mass and movement time between two groups. ***$P < 0.001$.

that female football players exhibited faster movement times compared to the control group, [t (24) = −3.369, P = 0.003, d = 1.325] (see Fig. 12C).

Furthermore, we also examined whether physiological signals in female football players are good indicators of their reaction times obtained from cognitive function tests. Therefore, we examined the association between physiological data measured by Garmin and reaction performance in the female football players group. Fig. 13 shows the correlation matrix between the 1-month heart rate characteristics measured by the Garmin wearable device and the cognitive function test scores (including reaction time and movement time) of the female football players group. The results showed that the movement time was negatively correlated with the median of average heart rate (r = −0.83, P = 0.011) and rest heart rate (r = −0.72, P = 0.042). Interestingly, the results also exhibited a negative correlation between reaction time and the median of average heart rate (r = −0.73, P = 0.042) in female football players. The results

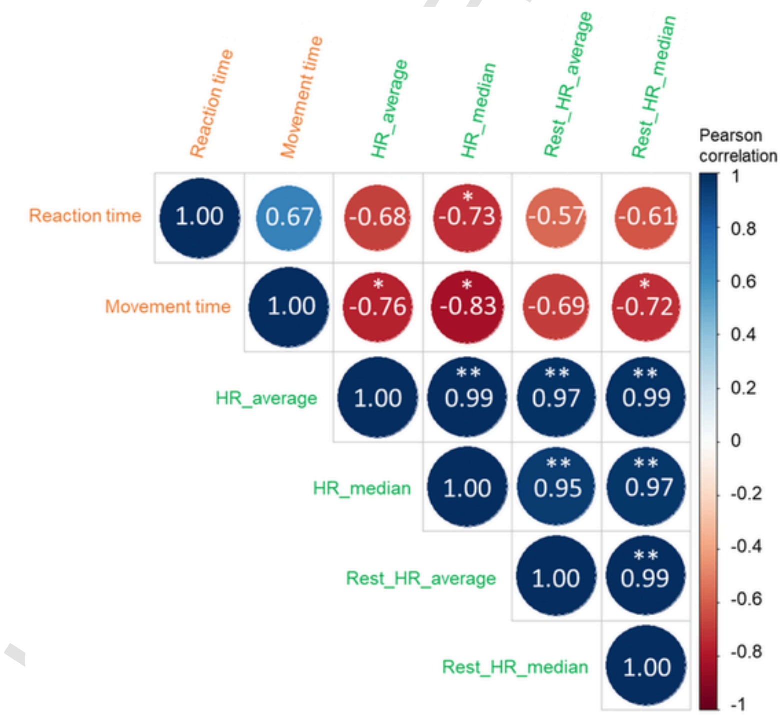


Fig. 13 The correlation matrix between reaction performance and heart rate related index from Garmin. HR = heart rate. **P < 0.01, *P < 0.05.

indicate that female football players' heart rate plays a regulatory role in their reaction time and movement performance.

On the other hand, the correlation between heart rate and movement time might change based on several variables, including stress levels, general health, and physical fitness, higher heart rates could be linked to heightened stress or arousal, resulting in faster movements under specific circumstances (Reimers et al., 2018). Previous studies has also found that football players exhibit different heart rate states depending on their playing positions (Ali and Farrally, 1991; Mills and Eglon, 2018). Ali and Farrally (1991) provided that the midfielders and forwards players often have greater average heart rates through games than defensive players. Thus, it's also crucial to determine whether heart rate management can improve cognitive function and result in beneficial outcomes from exercise training.

6. Discussion

With a multidimensional data analytical approach (including Garmin heart rate data, cognitive function scores, and body composition data), we observed that compared to female college students of the same age group, elite female football athletes exhibit higher limb muscle mass and shorter movement times in cognitive function scores. Additionally, there is a significant negative correlation between limb muscle mass and movement time, as well as between median heart rate and movement time. Thus, it is crucial to determine whether heart rate management can improve cognitive function and result in beneficial outcomes from exercise training in future studies.

Previous studies have used body composition analysis to predict football athletes' physical characteristics and body types, as well as to discuss the correlation with sports performance (Campa et al., 2020; Sebastiá-Rico et al., 2023). In this study, the measurement of body composition performance in female football athletes is consistent with previous studies, revealing that athletes have higher muscle strength, BMR, and body fat in specific areas compared to the control group. Interestingly, this study further provided that an inverse relationship between muscle mass and movement time was also discovered through an aggregated analysis of data from athletes and a control group of the same age range and gender. This suggests that greater muscle mass is linked to shorter movement times in female football athletes. Previous studies indicated that football athletes must exhibit fast and agility in order to improve sports performance during gameplay (Hasan, 2023; Peñailillo et al., 2016). Thus, increasing muscle strength among female football athletes may have beneficial effects on movement speed and agility.

Due to the athletes' long-term engagement in physical training, they have higher muscle mass compared to the control group, leading to shorter movement times. Additionally, after comparing the cognitive function between female soccer athletes and the general group, this study further examines the relationship between reaction performance and physiological status among female football athletes. By analyzing athletes' cognitive function scores (response time, movement time) and monthly long-term heart rate monitoring data, two trends were identified: (1) the median daily heart rate is inversely related to response time, and (2) the median and mean daily heart rates are inversely related to movement time.

Previous studies have mainly examined the acute effects of athletes' heart rate regulation during competitions on athletic performance (Matthew and Delextrat, 2009; Pueo et al., 2017). However, whether the average heart rate exhibited by athletes through prolonged training can also predict athletic performance is an important issue. This study found a negative correlation between the higher daily median heart rate and reaction time in female football athletes. The results reveal that female athletes with higher daily median heart rates demonstrate shorter reaction times. Interestingly, higher daily median heart rates also positively affect movement speed performance. This study indicate that cognitive function response time and movement time may be estimated from the daily median heart rate of female football athletes. For coaches, evaluating cognitive function performance based on daily heart rate data can serve as a quick reference for talent selection in coaching decisions.

However, due to the limited sample size of elite female football players (n = 14) with an average age of 20.6 ± 1.3 years in this study, the generalizability of the results is constrained to similar groups of players. Additionally, since this study primarily focused on elite female football players, the findings may not be applicable to other genders or sports disciplines. Future research could further validate and refine the findings by increasing the sample size and expanding the scope of the study population.

An integrative platform incorporated wearable devices, cognitive function tests, and body composition measurement into the training program can be beneficial for players and for coaching effectiveness. Heart rate management strategies and muscle mass enhancement training can potentially monitor athletes' daily performance and provide advices for improvement. The data platform also can capture the beat-to-beat-interval (BBI) signal from Garmin wearable devices and calculate HRV in time domain and frequency domain to monitor athletes' stress state. For instance, it has been shown that the High-Intensity Interval Training (HIIT) sessions that simultaneously stimulate the cardiovascular system and cognitive responses to dynamic game scenarios can optimize the balance between physical exertion and cognitive demands (Hsieh et al., 2021; Kao et al., 2021; Park et al., 2021; Taylor et al., 2022). The Garmin wearable device records the athletes' BBI during training in real-time, as well as the athletes' heart rate and sleep hours throughout the day. The body composition measurements were performed weekly and the cognitive function tests were performed monthly. Observing multidimensional data at different time scales provides coaches with a detailed athlete training profile, physical activities and status.

Using daily heart rate data to estimate the reaction time and movement time of football team athletes can improve the objective analyses of athletic competitions. It can also highlights the significance of empirical research in evaluating sports performance. For instance, the role of goalkeepers in intercepting shots promptly necessitates superior movement efficiency. By analyzing the daily heart rate of all team members, coaches can potentially select goalkeepers more accurately then making training and tactics more effective. This integrative research approach and its platform can offer a rigorous and evidence-based framework for assessing athletes' cognitive prowess and empowers coaches to make astute decisions in talent identification and training regimen development.

7. Conclusion

This study used the Python Flask framework with a MySQL database to establish the data platform. This platform serves as a comprehensive tool for collecting Garmin data, cognitive function scores, and body composition data. It also provides visualization interfaces for both athletes and coaches, allowing athletes to access their physiological and cognitive data via computers or mobile devices. This enhances athletes' ability to self-regulate training and improve training effectiveness. Additionally, it enables coaches to monitor the training intensity of the entire team.

The diverse data sources, such as Garmin data, cognitive function scores, and body composition data, are crucial for in-depth sports science data analysis. In the future, we will continue to add more data sources to enhance the diversity of our data platform. This expansion will aid in analyzing additional potential correlated data, ultimately achieving the goal of precision sports science.

Conflict of interest

We declare no competing finical interests.

Acknowledgments

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